Md. Tanvirul Islam Rifat

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About Me

BSc in Computer Science from BRAC University (2026) who builds complete, tested software: **full-stack web apps** (Flask, React, Node.js, MongoDB), **deep learning models** for medical imaging and sketch recognition, and **systems software** in C and C++. Open to roles in **software engineering, full-stack development, AI/ML** and **data science**.

Technical Skills

Programming	:	Python, C, C++, JavaScript, SQL
Frontend	:	React, HTML, CSS, Jinja2
Backend	:	Node.js, Express.js, Flask, FastAPI, SQLAlchemy, REST APIs, JWT Authentication
Database	:	MongoDB, MySQL, SQLite
ML & AI	:	PyTorch, Scikit-learn, Pandas, NumPy, SciPy, Matplotlib, CNNs, Vision Transformers, U-Net, Mamba
Systems	:	Linux, xv6 (RISC-V), Flex & Bison, OpenGL, Arduino, Cisco Packet Tracer
Tools	:	Git, GitHub, GitHub Actions (CI), Pytest, Make, Kaggle, Jupyter Notebook, VS Code
Core CS	:	Data Structures, Algorithms, OOP, DBMS, Operating Systems, Computer Networks, Compiler Design
Spoken	:	English (Professional), Bengali (Native), Spanish (A2), Chinese (A1)

Education

BRAC University — Dhaka, Bangladesh
BSc in Computer Science — *2022 – 2026* — *CGPA: 3.53 / 4.00*

Govt. Syed Hatem Ali College, Barishal — Barishal Board
HSC — Science — *2020* — *GPA: 5.00 / 5.00*

Barishal Zilla School — Barishal Board
SSC — Science — *2018* — *GPA: 5.00 / 5.00*

Research & Publications

HyMaC-Net — Hybrid Mamba–CNN for Medical Image Classification *PyTorch, Mamba* — BSc Thesis | GitHub

- **Accepted for publication** in *Scientific Reports* (Nature Portfolio), with S. Das, S. Chakraborty and S. H. Turja.
- A **22-step ablation** cut the model from **138M to 20.5M** parameters and raised PathMNIST from **89.0% to 94.0%**.
- On **12 medical benchmarks**, the 9.4M-parameter model matches MedViT-S on MedMNIST with **3.9× less compute** and beats the best compact baseline on Kvasir by **4.1 points**.

SketchEffNet — Efficient Sketch Recognition *PyTorch, GitHub Actions* — IEEE-format paper | GitHub

- Sole author. A **1.0M-parameter** CNN: **81.45%** on 50 QuickDraw classes, **11.2× fewer parameters** than ResNet18.
- Made every reported number checkable: **40 automated result checks** and **40 unit tests** run in CI on each push.

Projects

Pet Segmentation & Breed Classification *PyTorch, U-Net, ResNet34* — GitHub

- One network segments the pet and names its breed (37 classes); best of **6 models**: **0.93 mIoU**, **94.3%** accuracy.

TourPlanner — Travel Booking Website *Python, Flask, SQLAlchemy, SQLite* — GitHub

- Tour booking website with **three user roles** (traveller, hotel manager, travel agent) and a 7-table database.

SimpleFS — File System in C *C, Python, Make, GitHub Actions* — GitHub

- In a team of two, wrote a block-based file system with inodes and bitmaps, a Python checker and a **38-test suite** in CI.

Mini C Compiler *C++, Flex, Bison* — GitHub

- Compiler for a C subset: lexer, parser, scoped symbol table, type checking and **three-address code** generation.

QuakeNet — Disaster-Response Network *Cisco Packet Tracer, Python* — GitHub

- In a team of four, built a 7-router network (VLSM, RIPv2, IP SLA failover) and a verifier that traced **112 flows**.

Leadership & Awards

- **Senior Executive, BRAC University Computer Club** (Dec 2023 – Dec 2025): organized campus-wide technology events, including programming contests and Database Wizards.
- **Bronze Standard**, The Duke of Edinburgh's International Award.